

Dexamethasone and risk of nausea and vomiting and postoperative bleeding after tonsillectomy in children: a randomized trial.

SCIFACT-EVIDENCE-23349986

Evidence abstract

CONTEXT Dexamethasone is widely used to prevent postoperative nausea and vomiting (PONV) in pediatric tonsillectomy.

OBJECTIVE To assess whether dexamethasone dose-dependently reduces the risk of PONV at 24 hours after tonsillectomy.

DESIGN, SETTING, AND PATIENTS Randomized placebo-controlled trial conducted among 215 children undergoing elective tonsillectomy at a major public teaching hospital in Switzerland from February 2005 to December 2007.

INTERVENTIONS Children were randomly assigned to receive dexamethasone (0.05, 0.15, or 0.5 mg/kg) or placebo intravenously after induction of anesthesia.

Acetaminophen-codeine and ibuprofen were given as postoperative analgesia.

Follow-up continued until the 10th postoperative day.

MAIN OUTCOME MEASURES The primary end point was prevention of PONV at 24 hours; secondary end points were decrease in the need for ibuprofen at 24 hours and evaluation of adverse effects.

RESULTS At 24 hours, 24 of 54 participants who received placebo (44%; 95% confidence interval [CI], 31%-59%) had experienced PONV compared with 20 of 53 (38%; 95% CI, 25%-52%), 13 of 54 (24%; 95% CI, 13%-38%), and 6 of 52 (12%; 95% CI, 4%-23%) who received dexamethasone at 0.05, 0.15, and 0.5 mg/kg, respectively ($P < .001$ for linear trend).

Children who received dexamethasone received significantly less ibuprofen.

There were 26 postoperative bleeding episodes in 22 children.

Two of 53 (4%; 95% CI, 0.5%-13%) children who received placebo had bleeding compared with 6 of 53 (11%; 95% CI, 4%-23%), 2 of 51 (4%; 95% CI, 0.5%-13%), and 12 of 50 (24%; 95% CI, 13%-38%) who received dexamethasone at 0.05, 0.15, and 0.5 mg/kg, respectively ($P = .003$).

Dexamethasone, 0.5 mg/kg, was associated with the highest bleeding risk (adjusted relative risk, 6.80; 95% CI, 1.77-16.5).

Eight children had to undergo emergency reoperation because of bleeding, all of whom had received dexamethasone.

The trial was stopped early for safety reasons.

CONCLUSION In this study of children undergoing tonsillectomy, dexamethasone decreased the risk of PONV dose dependently but was associated with an increased risk of postoperative bleeding.

TRIAL REGISTRATION clinicaltrials.gov Identifier: NCT00403806.

Document metadata

Corpus | SciFact

Document ID | 23349986

Evidence checklist

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16. **TRIAL REGISTRATION** [clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT00403806) Identifier: NCT00403806.

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